

第九周作业参考解答

秦政

2022 年 11 月 2 日

目录

1 2022.10.24 第十四次课作业	1
1.1 课本 $P_{372}T_{11}$	1
1.2 课本 $P_{372}T_{12}$	4

1 2022.10.24 第十四次课作业

1.1 课本 P₃₇₂T₁₁

在等温大气中，密度由 $\bar{\rho} = \rho_0 e^{-2\beta z}$ 给出，其中 z 是铅垂方向的坐标。若用 w 表示速度的铅垂分量，证明圆频率为 σ 的微波动由

$$\left[\left(1 - \frac{N^2}{\sigma^2} \right) \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + \frac{\partial^2}{\partial z^2} - 2\beta \frac{\partial}{\partial z} \right] w = 0$$

给出，其中

$$N^2 = -\frac{g}{\bar{\rho}} \frac{d\bar{\rho}}{dz} = 2\beta g,$$

x, y, z 是笛卡尔坐标。（注：默认在未说明情况下，就是不可压缩的）

参考解答：

等温大气中的任何量 X ，都可以写成它的平均量（不含时）和它的扰动量（小振动）之和

$$X = \bar{X} + X'$$

本题要求解 w 的微波动，即 w' ，接下来的过程中 w' 就是本题中的 w 。由平均量与时间无关，铅垂方向有

$$\text{一般方程: } \frac{\partial w}{\partial t} = \frac{\partial w'}{\partial t} = -g - \frac{1}{\bar{\rho}} \frac{\partial p}{\partial z}$$

$$\text{压高关系: } \frac{\partial \bar{\rho}}{\partial z} = -\bar{\rho} g$$

$$\frac{\partial w'}{\partial t} = -g - \frac{1}{\bar{\rho}} \frac{\partial p}{\partial z} = -g - \frac{1}{\bar{\rho} + \rho'} \frac{\partial (\bar{p} + p')}{\partial z} = -g - \frac{1}{\bar{\rho} \left(1 + \frac{\rho'}{\bar{\rho}} \right)} \frac{\partial (\bar{p} + p')}{\partial z}$$

$$= -g - \frac{1}{\bar{\rho} \left(1 + \frac{\rho'}{\bar{\rho}} \right)} \left(-\bar{\rho} g + \frac{\partial p'}{\partial z} \right)$$

$$= -g - \frac{1}{\bar{\rho}} \left(1 - \frac{\rho'}{\bar{\rho}} \right) \left(-\bar{\rho} g + \frac{\partial p'}{\partial z} \right)$$

去除高阶小量后

$$\frac{\partial w'}{\partial t} = -\frac{\rho'}{\bar{\rho}} g - \frac{1}{\bar{\rho}} \frac{\partial p'}{\partial z} \quad (1)$$

小振动的运动方程组

$$\begin{aligned}\frac{\partial u'}{\partial t} &= -\frac{1}{\bar{\rho}} \frac{\partial p'}{\partial x} \\ \frac{\partial v'}{\partial t} &= -\frac{1}{\bar{\rho}} \frac{\partial p'}{\partial y} \\ \frac{\partial w'}{\partial t} &= -\frac{\rho'}{\bar{\rho}} g - \frac{1}{\bar{\rho}} \frac{\partial p'}{\partial z}\end{aligned}$$

对 u' 、 v' 分别进行 x 、 y 偏导，以得到要证明式子的 $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)$ 形式

$$\frac{\partial}{\partial t} \left(\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} \right) = -\frac{1}{\bar{\rho}} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) p'$$

由大气不可压缩特性

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0$$

则

$$\frac{\partial}{\partial t} \left(-\frac{\partial w'}{\partial z} \right) = -\frac{1}{\bar{\rho}} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) p'$$

上式进行对 z 求偏导，以得到要证明式子的 $\frac{\partial^2}{\partial z^2}$ 项，则上式右边需要对 $\bar{\rho}$ 和 p' 都进行偏导

$$\begin{aligned}\frac{\partial}{\partial t} \frac{\partial^2 w'}{\partial z^2} &= \frac{1}{\bar{\rho}} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \frac{\partial p'}{\partial z} + \left(-\frac{1}{\bar{\rho}^2} \right) \frac{d\bar{\rho}}{dz} \cdot \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) p' \\ &= \frac{1}{\bar{\rho}} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \frac{\partial p'}{\partial z} + \frac{N^2}{g} \left(-\frac{\partial}{\partial x} \frac{\partial u'}{\partial t} - \frac{\partial}{\partial y} \frac{\partial v'}{\partial t} \right) \\ \frac{\partial}{\partial t} \frac{\partial^2 w'}{\partial z^2} &= \frac{1}{\bar{\rho}} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \frac{\partial p'}{\partial z} + \frac{N^2}{g} \frac{\partial}{\partial t} \left(\frac{\partial w'}{\partial z} \right) \quad (2)\end{aligned}$$

不可压缩大气连续方程

$$\begin{aligned}\frac{d\rho}{dt} &= 0 \quad (\text{注意这里一定是 } \rho \text{ 不是 } \bar{\rho}) \\ \frac{d\rho}{dt} &= \frac{\partial \rho'}{\partial t} + (\bar{u} + u') \frac{\partial (\bar{\rho} + \rho')}{\partial x} + (\bar{v} + v') \frac{\partial (\bar{\rho} + \rho')}{\partial y} + (\bar{w} + w') \frac{\partial (\bar{\rho} + \rho')}{\partial z}\end{aligned}$$

去除平均项和高阶小量后

$$\begin{aligned}\frac{d\rho}{dt} &= \frac{\partial \rho'}{\partial t} + w' \frac{\partial \bar{\rho}}{\partial z} = 0 \Rightarrow \frac{\partial}{\partial t} \left(\frac{\rho'}{\bar{\rho}} \right) + w' \frac{1}{\bar{\rho}} \frac{\partial \bar{\rho}}{\partial z} = 0 \\ \frac{\partial}{\partial t} \left(\frac{\rho'}{\bar{\rho}} \right) - \frac{N^2}{g} w' &= 0 \quad (3)\end{aligned}$$

利用 (3)

$$\frac{\partial}{\partial t} \left(\frac{\rho'}{\bar{\rho}} \right) = \frac{N^2}{g} w'$$

(1) 式进行对 t 求偏导后代入 (3)

$$\frac{\partial^2 w'}{\partial t^2} + g \frac{\partial}{\partial t} \left(\frac{\rho'}{\bar{\rho}} \right) + \frac{\partial}{\partial t} \left(\frac{1}{\bar{\rho}} \frac{\partial p'}{\partial z} \right) = 0 \Rightarrow \frac{\partial^2 w'}{\partial t^2} + N^2 w' + \frac{\partial}{\partial t} \left(\frac{1}{\bar{\rho}} \frac{\partial p'}{\partial z} \right) = 0$$

再对上面式子进行 $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right)$ 算符运算, 即可得

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \frac{\partial^2 w'}{\partial t^2} + N^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) w' + \frac{\partial}{\partial t} \left[\frac{1}{\bar{\rho}} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \frac{\partial p'}{\partial z} \right] = 0$$

代入 (2)

$$\begin{aligned} \frac{1}{\bar{\rho}} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \frac{\partial p'}{\partial z} &= \frac{\partial}{\partial t} \frac{\partial^2 w'}{\partial z^2} - \frac{N^2}{g} \frac{\partial}{\partial t} \left(\frac{\partial w'}{\partial z} \right) \\ \left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \frac{\partial^2}{\partial t^2} + N^2 \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + \frac{\partial^2}{\partial t^2} \frac{\partial^2}{\partial z^2} - \frac{N^2}{g} \frac{\partial^2}{\partial t^2} \frac{\partial}{\partial z} \right] w' &= 0 \end{aligned}$$

下面将方程代入 w' 是圆频率为 σ 的微波动的条件, 可设

$$w' = W e^{i(kx+ly+mz-\sigma t)} \Rightarrow \frac{\partial^2}{\partial t^2} w' = -\sigma^2 w'$$

则关于 w' 方程可以写为

$$\left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (\sigma^2 - N^2) + \sigma^2 \frac{\partial^2}{\partial z^2} - 2\beta \sigma^2 \frac{\partial}{\partial z} \right] w' = 0$$

w' 即为本题中的 w

$$\left[\left(1 - \frac{N^2}{\sigma^2} \right) \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) + \frac{\partial^2}{\partial z^2} - 2\beta \frac{\partial}{\partial z} \right] w = 0$$

PS: 这道题难度较大, 主要是考察运动方程量级分析及去除高阶小量, 量级分析可以让复杂的运动方程简化。注意在运用方程时还是应该先写出一一般方程再根据量级简化 (如题中运用连续方程时, 不能直接对密度平均量写连续方程)。

1.2 课本 P₃₇₂T₁₂

如果沿全渠道各个截面相等，截面面积为 S ，并且水面上的截面宽度为 b ，求长波在渠道中的传播速度.

参考解答：

长波运动的基本方程组

$$\begin{aligned}\frac{\partial u}{\partial t} &= -g \frac{\partial \zeta}{\partial x} \\ \frac{\partial \zeta}{\partial t} + \frac{\partial(uh)}{\partial x} &= 0\end{aligned}$$

全渠道各个截面相等，则深度 h 为常数， u 满足的方程为

$$\frac{\partial^2 u}{\partial t^2} - gh \frac{\partial^2 u}{\partial x^2} = 0$$

令 $c = \sqrt{gh}$ ，得到 u 通解

$$u = c[F(x - ct) + f(x + ct)]$$

长波传播速度为

$$c = \sqrt{gh} = \sqrt{g \frac{S}{b}}$$

PS：这是长波的一个简单运用，大家可以复习一下学过的几种波动。